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Originality declaration

I, [**Ruijue Song**], confirm that the work presented in this assessment is my own. Where information has been derived from other sources, I confirm that this has been indicated in the work.

I used Gemini to help me with data analysis, and I will specifically mark the places where I used it.

And I have extra time of Reasonable Academic Adjustments.

Initial project scope

Background

The gambling market in the UK is substantial, generating £14.2 billion in 2020. The associated gambling-related harms are increasingly viewed as a critical public health issue rather than merely a leisure activity.

In response to the growing scale of gambling harms, the UK Government published the White Paper **"High stakes: gambling reform for the**

digital age" in 27 April 2023. This White Paper prioritized protecting vulnerable people (e.g., children and young adults) from gambling harms. Despite the legislative focus and the gambling-related harm review in 2023, the spatial dimension of gambling harms is not taken into consideration, such as whether the betting stores (bookmakers) physically cluster in vulnerable communities.

Consequently, the bookmakers dataset extracted from OSM data provides a critical window into the spatial distribution patterns. Acting as a consultant of Greater London Authority (GLA), this report seeks to understand what are the spatial patterns of gambling density in London, thereby providing the missing spatial analysis to inform local licensing and planning decisions.

Research question and hypothesis

Research question: What are the spatial patterns of betting stores(bookmakers) in London in 2025? Do they form statistically significant high-density zones similar to one another (hotspots)?

Hypothesis:

- H_1 (Alternative Hypothesis): Betting stores in London in 2025 are not randomly distributed but exhibit significant spatial clustering, with certain districts experiencing disproportionately high density compared to others. And there are hotspot zones in London.

- H_0 (Null Hypothesis): Betting stores in London in 2025 are randomly distributed without showing any hotspot zones.

Evaluation of research datasets

dataset: Bookmakers dataset extracted from OSM data

- **Source & Nature**: Provide by OSM data, the dataset contains polygons of bookmakers locations in London, which has to be transformed into point data for following density calculation.
- **CRS**: The raw data is in WGS 84 (EPSG:4326). A key pre-processing step will involve transforming this to British National Grid (EPSG:27700) to enable valid metric spatial operations (e.g., calculating density per km^2).
- **Collection Method**: The dataset is extracted from **OpenStreetMap (OSM)**, which is a collaborative project where data is created and maintained by volunteers. Unlike official administrative registers (e.g., from the UK Gambling Commission), this dataset relies on **crowdsourced contributions**. As a crowdsourced dataset, it is subject to contributor bias and temporal lag. There is no guarantee that every betting store in London has been mapped. Newly opened stores could be missing and permanently closed shops may not have been deleted.
- **Representation**: OSM coverage is often better in densely

populated or affluent areas where more active contributors reside. Consequently, betting shops in deprived peripheral areas might be slightly underrepresented.

Explanation of the data wrangling to undertake

1. ****CRS Transforming:**** All spatial datasets (OSM bookmaker points, MSOA boundaries, London boundary) will be transformed to the British National Grid (EPSG: 27700), which is essential for accurate area calculations (required for density) and spatial distance measurements, avoiding distortions inherent in the raw WGS84 geographic system.
2. ****Data Cleaning:**** Remove the duplicates in bookmaker points dataset and select the related columns for following analysis.
3. ****Spatial Subsetting:**** The OSM bookmaker points will be filtered to retain only those falling strictly within the study area (London MSOAs boundaries). This step is necessary to remove extraneous data points located in the corners of the bounding box, ensuring the analysis is strictly confined to the GLA jurisdiction.
4. ****Data Aggregation:**** This is the most critical step. Betting shop points will be spatially joined to the MSOA polygons to calculate a raw count per spatial unit and these counts will be converted to density. Normalising by area is vital to mitigate the Scale Effect of the Modifiable Areal Unit Problem (MAUP). Without this,

physically larger MSOAs in outer London might artificially seem to have more gambling issues simply due to their size.

Justified plan for the spatial data analysis that will be required

1. **Kernel Density Estimation (KDE): Visualising Intensity**

We firstly generate a KDE surface to from the cleaning betting shop points to identify the intensity in London, which could provide us with a initial understanding of bookmakers distribution and intensity.

2. **Global Moran's I: Testing for Global Clustering in London**

Following data aggregation to MSOAs, the Global Moran's I statistic will be calculated for betting stores density.

Visual patterns in KDE can be vague and misleading. To rigorously answer the research question, we must statistically test against the Null Hypothesis of Complete Spatial Randomness (CSR). Global Moran's I is the appropriate statistic for areal data (MSOAs) to quantify whether there is clustering across the entire city, confirming if the observed patterns are systematic rather than random noise.

3. **Getis Ord G_i^* : Identifying Local Hotspots**

While Global Moran's I confirms if clustering exists in London, it cannot tell us where. Getis Ord G_i^* is specifically selected over

other local indicators (like Local Moran's I) because it explicitly identifies "High-High" clusters (high density surrounded by high density). This directly answers the second part of the research question: "where are significant hotspots?". It also provides actionable evidence for identifying specific neighbourhoods where gambling availability has reached saturation levels, potentially requiring policy intervention.

Constraints & Limitations

1. Data Reliability

The betting shop data is sourced from OpenStreetMap (OSM), which is Volunteered Geographic Information. It may suffer from completeness bias, where shops in less active contributor areas (often peripheral or deprived) are under-mapped, potentially leading to an underestimation of the density in the very areas of interest.

2. The Modifiable Areal Unit Problem (MAUP)

The analysis relies on aggregating points to MSOAs. The results (e.g., identified hotspots) are dependent on these specific administrative boundaries. Using different scales (e.g., boroughs) could yield different clustering patterns or spatial correlations.

Data Pre-processing

```
``{r}
```

```
library(tidyverse)
```

```
library(sf)
```

```
library(tmap)
```

```
library(ggplot2)
```

```
library(janitor)
```

```
library(osmdata)
```

```
library(spatstat) # Point Pattern Analysis (KDE)
```

```
library(spdep) # Spatial statistics (Global Moran's I and Local Getis Ord Gi)
```

```
...
```

Data Loading

1. Loading the MSOA boundary data Due to GitHub's file size limits ($>50\text{MB}$), the raw MSOA boundary data is not included in this repository.

URL of MSOA boudnary:

<https://geoportal.statistics.gov.uk/datasets/3b3b1cb15a744855881b1489741b861b_0/about>

```
``{r}
```

```
MSOAs <-
```

```
st_read("data/gambling_related_harms_data/Middle_layer_Super_Output_Areas_December_2021_Boundaries_EW_BFE_V8_-1517080999235121072/MSOA_2021_EW_BFE_V8.shp") %>%
```

```
st_transform(., 27700) #Ensure it is in British National Grid
```

```
st_crs(MSOAs) #Check its crs
```

```
...
```

2. Loading London boundary data

URL:

```
<https://data.london.gov.uk/download/20od9/114d1137-e339-4b50-b409-124c17f4b59a/gla.zip>
```

```
``{r}
```

```
London <-
```

```
st_read("data/gambling_related_harms_data/gla/London_GLA_Boundary.shp") %>%
```

```
  st_transform(., 27700)
```

```
qtm(London)
```

```
...
```

Subsetting spatial units (MSOAs) to a study area (London)

```
``{r}
```

```
#centroids of MSOAs across UK
```

```
MSOAs_centroids <- st_centroid(MSOAs)
```

```
#centroids within the London boundary. London is an sf object of the Greater London boundary
```

```
London_MSOAs <- MSOAs[st_within(MSOAs_centroids, London, sparse = FALSE), ]
```

```
qtm(London_MSOAs)
```
```

### 3. Extracting the bookmakers dataset from OSMs

```
```{r}

Bounding_box <- getbb("London")

#check it worked

Bounding_box

#get the OSM data within the bounding box

OSM_bookmaker <- Bounding_box %>%
  opq() %>%
  add_osm_feature(key = "shop", value = "bookmaker")%>%
  osmdata_sf()

#extract the polygons of bookmakers

bookmakers<-OSM_bookmaker$osm_polygons

#convert to points

bookmaker_points <- st_centroid(bookmakers)

```

```{r}

# CRS transforming for bookmaker points to British National Grid

bookmaker_points <- bookmaker_points %>%
  st_transform(., 27700)
```

```
st_crs(bookmaker_points)
```

```
...
```

```
``{r}
```

```
summary(bookmaker_points)
```

```
...
```

Let us have a look at the points distribution firstly

```
``{r}
```

```
tmap_mode("plot")
```

```
# Set the base map
```

```
tm_shape(London_MSOAs)+
```

```
  tm_polygons(fill_alpha = 0.3)+
```

```
# Set the bookmakers dots
```

```
tm_shape(bookmaker_points)+
```

```
  tm_dots(fill = "blue", size = 0.2)
```

```
...
```

It is obvious that some points fall out of the London boundary, and we have to remove them by subsetting in data wrangling part.

```
``{r}
```

```
#chek the column names and datatype list of bookmarker points data
```

```
Datatypeslist <- bookmaker_points %>%
```

```

st_drop_geometry() %>%
summarise_all(class) %>%
pivot_longer(everything(),
              names_to = "All_variables",
              values_to = "Variable_class")

```

Datatypes

...

Inspection of the dataset structure reveals that a significant number of attribute columns contain NA values. As these attributes (e.g., address details, opening hours) are not required for the proposed spatial density analysis, the dataset will be streamlined.

In the data wrangling stage, we will select only the relevant columns (e.g., name or osm_id) and the essential geometry column. This reduces computational overhead and ensures a clean dataset for aggregation.

Data Wrangling

1. Data Cleaning: remove duplicates and select related columns

```
``{r}
```

```
# First we remove the duplicates in bookmaker points
```

```
bookmaker_points <- bookmaker_points %>%
```

```
  clean_names() %>%
```

```
  distinct()
```

```
# Then we keep only columns relevant to analysis: osm_id, name and geometry

bookmaker_selected <- bookmaker_points %>%

  select(osm_id, name, geometry) %>%

  # This step is to filter out rows with empty geometry (and I asked Gemini how to do it
  # with sf)

  # Although it is rare in raw OSM data

  # I still take this step to ensure that subsequent spatial operations do not fail due to
  # corrupt vector data

  filter(!st_is_empty(geometry))

...

```

2. Spatial Subsetting

```
```{r}

We remove any points falling outside London

bookmaker_sub <- bookmaker_selected[London_MSOAs,]

...

```

#### #### 3. Exploratory Mapping: bookmaker points in London

```
```{r}

# We plot to double check whether the points out of London removed

tmap_mode("plot")

tm_shape(London_MSOAs)+

  tm_polygons(fill_alpha = 0.3)+

tm_shape(bookmaker_sub)+

```

```
tm_dots(fill = "blue", size = 0.2)
...

```

4. Data Aggregation

Next, we calculate Density (count of betting stores per unit of area).

This reveals how concentrated the gambling is.

```
```{r}
points_sf_joined <- London_MSOAs %>%
 clean_names() %>%
 # First count points inside polygons
 # st_intersects returns a list of points found in each polygon
 # lengths() counts how many points in each unit
 mutate(n = lengths(st_intersects(., bookmaker_sub))) %>%
 # Calculate the area of each spatial unit
 mutate(area = st_area(.)) %>%
 # Calculate the density
 # We multiply the density by 1,000,000 to convert it from per 1/m^2 to 1/km^2
 # Cause at first I used 1/m^2 to plot, find the density is so small (e.g., 0.000000008)
 # that is hard to read
 mutate(density = (n/area)*1000000) %>%
 # Select related variables
 select(msoa21cd, msoa21nm, density, n)
...

```

We plot a choropleth map of betting stores density in London to perform

a preliminary visual inspection for potential 'saturation zones' or clustering tendencies that warrant further investigation using global and local spatial statistics.

```
```{r}

points_sf_joined <- points_sf_joined %>%

  group_by(msoa21cd) %>%

  summarise(density = first(density), msoa21nm = first(msoa21nm), bookmaker_count
= first(n))

# Plot the choropleth map of bookmaker density

tm_shape(points_sf_joined) +

  tm_polygons(fill = "density",

              fill.scale = tm_scale_intervals(

                values = "brewer.blues",

                style = "jenks"),

              fill.legend = tm_legend(

                title = "Bookmaker Density in London",

                title.size = 0.85,

                size = 0.8,

                poisson = tm_pos_out(

                  "right",

                  "center",

                  pos.v = "center"

                )

              ))

```
```

The map rejects a random distribution, revealing a distinct core-periphery pattern with the highest densities ( $> 5.9$  shops/ $\text{km}^2$ ) concentrated in Inner London. Significant 'islands' of high density appear in suburban hubs (e.g., Croydon, Stratford)-I asked Gemini to identify their names because I am not familiar with London districts. These potential 'saturation zones' (approaching 9 shops/ $\text{km}^2$ ) suggest strategic clustering in high-footfall areas. In the context of the Public Health England review, these 'saturation zones' represent areas of hyper-accessibility, where the risk of gambling-related harm is likely elevated due to the concentrated volume of betting opportunities.

While visually suggesting clustering, visual inspection is prone to subjectivity. Points pattern and spatial autocorrelation analysis are required to statistically confirm if these patterns are significant.

## ## Data Analysis

### ### Points Pattern Analysis

#### #### Kernel Density Estimation (KDE)

**\*\*Why KDE is appropriate while Ripley's K is not:\*\*** While Ripley's K is valuable for quantifying the existence and scale of clustering (Global Statistics), it does not identify the location of these clusters. KDE shows us WHERE the highest density of betting stores is, which allows for the precise geographic identification of 'risk environments' where

gambling availability is highest. Ripley's K only indicates the scale of clustering instead of its specific location.

### 1. Defining Observation Window (London boundary)

```
```{r}
# Set the London boundary as the window
window <- as.owin(London)
plot(window)
```
```

- ### 2. Data Conversion (sf $\rightarrow$ ppp) The sf point object (bookmakers) cannot be used directly for density calculation. It must be converted into a Planar Point Pattern (ppp) object, which is the specific format required by the spatstat package.

```
```{r}
# Extract the coodinations of bookmakers
coords <- st_coordinates(bookmaker_sub)

# Converting sf to ppp
bookmaker_sub.ppp <- ppp(x = coords[,1],
                        y = coords[,2],
                        window = window)
```
```

### 3. KDE

```

```{r}

# Run KDE

bookmaker_sub.ppp %>%

  density(., sigma = 1000) %>%

  plot()

```

```

This KDE surface reveals the continuous 'atmosphere' of gambling availability. It identifies a 'hyper-saturated' core (Yellow) and distinct polycentric 'flares' in Outer London (Red/Orange). These peripheral hotspots (e.g., Croydon, Stratford) confirm that shops are not randomly scattered but strategically anchored to high-footfall commercial arteries.

### ### Spatial Autocorrelation

#### #### Global Moran's I

**\*\*Why Global Moran's I is appropriate:\*\*** We use Global Moran's I to evaluate whether the spatial pattern of betting store density **\*\*across London (Global)\*\*** is clustering. A **\*\*Queen's Contiguity\*\*** spatial weight matrix was used to conceptualise spatial relationships because it captures all touching neighbours (sharing edges or corners). This is particularly appropriate for modelling 'risk environments' in London's irregular urban fabric, ensuring that the spatial spillover of gambling availability across complex administrative intersections is fully

accounted for.

## 1. Defining Wight Matrix: Queen Contiguity

```
```{r}
```

```
# Create Neighbor List (Queen Contiguity)
```

```
nb <- points_sf_joined %>%
```

```
  poly2nb(., queen = T)
```

```
# Create Spatial Weights List
```

```
lw <- nb %>%
```

```
  nb2listw(., style = "W", zero.policy = TRUE) # zero.policy=TRUE handles districts  
(islands) with no neighbors
```

```
```
```

## 2. Global Moran's I

```
```{r}
```

```
# Run Global Moran's I test
```

```
moranI <- points_sf_joined %>%
```

```
  pull(density) %>%
```

```
  as.vector() %>%
```

```
  moran.test(., lw)
```

```
moranI
```

```
```
```

The Global Moran's I test yields a Moran's I statistic of 0.165 with a Z-score (standard deviate) of 9.02 and a p-value of  $< 2.2e-16$ , revealing a positive spatial autocorrelation (clustering) and rejecting the Null Hypothesis (CSR).

The exceptionally high Z-score (9.02) confirms that the observed clustering is not random noise, but a structurally significant, systematic phenomenon across London. It provides statistical evidence supporting the 'predatory clustering' hypothesis, suggesting operators systematically co-locate to create continuous 'risk environments' rather than distributing randomly. To translate this city-wide finding into actionable policy intelligence, a Local Getis-Ord  $G_i^*$  analysis is required to pinpoint exactly where these hotspots.

#### Local Getis Ord  $G_i^*$

**Why Getis Ord  $G_i^*$  is appropriate:** We choose Local Getis Ord  $G_i^*$  over Local Moran's I (LISA) as the study prioritizes identifying 'hotspot zones' rather than spatial outliers. Informed by the Public Health England review,  $G_i^*$  specifically detects hotspots (high density surrounded by high density), which directly corresponds to contiguous 'risk environments' of hyper-accessibility. Coordinated policy intervention is needed in these areas.

1. Local Getis Ord  $G_i^*$

{}<sub>r</sub>

```
Run Local Getis Ord Gi* test
```

```
Gi <- points_sf_joined %>%
```

```
pull(density) %>%
```

```
as.vector()%>%
```

localG(., lw)

head(Gi)

///

## 2. Plot

 $\{\{r\}$ 

```
Add Gi data to points_sf_joined for plot
```

```
points_sf_joined <- points_sf_joined %>%
```

```
mutate(density_G = as.numeric(Gi))
```

```
Plot the Gi density map
```

```
tm_shape(points_sf_joined) +
```

tm\_polygons(

```
fill = "density_G",
```

```
fill.scale = tm_scale_intervals(values = "brewer.blues",
```

```
midpoint = "NA",
```

```
style = "jenks"),
```

```
fill.legend = tm_legend(title="Gi*, Bookmaker Density in London",
```

```
title.size=0.85,
```

size=0.8,

```
position=tm_pos_out("right",
```

...

Aligning with the Public Health England review, these High-High clusters represent "frontline risk environments". This validates the "predatory clustering" hypothesis, suggesting operators aggressively concentrate supply to maximize cumulative impact. For the GLA, this provides the specific evidentiary basis to designate "Cumulative Impact Zones" and refuse new licenses in these saturated districts.

#### #### Policy Implications

Targeting 'Hotspot Zones, the spatial analysis successfully translates the abstract concept of "gambling harm" into a concrete instrument for policy intervention. Rather than applying a blanket "one-size-fits-all" restriction, **policy should be spatially targeted**. In these

high-density hotspots, the policy should be stricter. New license applications should be refused because these areas already have enough betting stores. This would help stop the clustering that leads to higher risks of gambling harm in these communities.

#### #### Reflection on Datasets and Methods

1. **\*\*Data Issue:\*\*** The betting shop data comes from OpenStreetMap, which is created by volunteers. This might have sampling bias. For example, shops in poorer or outer areas might be missing because fewer volunteers map there.
2. **\*\*Methodological Limits (MAUP):\*\*** The results depend on the boundaries we used (MSOAs). This is the Modifiable Areal Unit Problem (MAUP). If we used different zones (like Boroughs), the results might look different.